Laundry

Imagine an unreliable clothes dryer that does one of two things when you put money in:

- It takes 20 minutes to dry your clothes. This event occurs with probability 0.6.
- It buzzes for 4 minutes and does nothing to your clothes. This even occurs with probability 0.4.

You plan to keep putting money into the machine until it dries your clothes. Assume that successive trials with the dryer are independent.

1. Let N be the random variable that counts how many times you have to run the dryer until it dries your clothes. What distribution does N have? Use the "~" notation and the name of the distribution.

2. The random variable N – 1 counts the number of times that the dryer fails before its first successful run. Therefore, 4(N – 1) is the amount of time the dryer spends on unsuccessful runs. Calculate the expected time E[4(N – 1)].

3. What is the expected time to dry your clothes?